



Machine Learning for TSP Free-Acid Monitoring

Forecasting, virtual sensing, and an advisory OPC UA deployment
prototype

Prepared by: **Ikram Benfella**

Host organization: OCP Group

Academic supervisor: El OTEBY Samir

Industrial supervisor: IMELLOUL M'hamed

Academic year: 2026–2027

June - September 2026

Contents

List of Abbreviations	v
1 Introduction	1
2 Company Presentation	2
2.1 OCP Group	2
2.2 Industrial activities relevant to the project	2
2.3 Relevant technical roles	3
3 Internship Context and Problem Definition	4
3.1 TSP monitoring context	4
3.2 Evolution of the work	4
4 Data Understanding and Preparation	6
4.1 Data source and scope	6
4.2 Variables and measurement semantics	6
4.3 Data quality and preprocessing	7
4.4 Feature engineering	7
5 Machine Learning Methodology	9
5.1 Problem formulation	9
5.2 Chronological validation	9
5.3 Models investigated	10
6 Experimental Results and Model Selection	11
6.1 Direct future-level benchmark	11
6.2 Target-anchored delta forecast	11
6.3 Target-free virtual sensor and stability research	12
7 Model Interpretability	14
8 Deployment Architecture	16
8.1 Architecture and safety boundary	16

8.2	Containers, services and configuration	16
9	Real-Time Processing and Deployment Implementation	18
9.1	Stage 1 CSV replay	18
9.2	Streaming pipeline and quality controls	18
9.3	Warm-up and supported modes	18
9.4	Heartbeat, watchdogs and recovery	19
10	Discussion	20
11	Contribution and Lessons Learned	21
12	Conclusion and Future Work	22
A	Evidence and Deployment Scope	24
A.1	Notebook and implementation evidence	24
A.2	Terms that require supervisor confirmation	24
A.3	Software verification scope	24

List of Figures

- 4.1 Leakage-controlled feature-preparation pipeline generated by Notebook 02. . . . 8
- 6.1 Delta forecasting performance, ablation and distribution-shift diagnostic. . . . 12
- 6.2 Target-free virtual-sensor comparison and diagnostics from Notebook 05. . . . 13
- 7.1 Grouped predictive importance and horizon stability from the direct-benchmark analysis. 15
- 8.1 Advisory information flow implemented and planned for industrial integration. . 16

List of Tables

- 3.1 Operational prediction contracts evaluated in the project. 4
- 4.1 Principal input families in the feature registry. 7
- 6.1 Direct future-level Ridge results on the held-out January period. 11
- 6.2 Target-anchored delta-Ridge test performance after future-level reconstruction. . 12
- 9.1 Operational states exposed by the inference service. 19

List of Abbreviations

Abbreviation	Meaning
CSV	Comma-Separated Values
DCS	Distributed Control System
HMI	Human-Machine Interface
JITL	Just-In-Time Learning
KNN	K-Nearest Neighbours
MAE	Mean Absolute Error
ML	Machine Learning
OPC UA	Open Platform Communications Unified Architecture
PLC	Programmable Logic Controller
RMSE	Root Mean Squared Error
SHAP	SHapley Additive exPlanations
TSP	Triple Super Phosphate
WAL	Write-Ahead Logging

Chapter 1

Introduction

The internship addressed a practical question in TSP production: whether historical, one-minute process measurements can support useful estimates or short-horizon forecasts of `SLURRY_FREE_ACID`. The work was developed as an industrial machine-learning project rather than as a single of-line model. It therefore included data investigation, leakage-controlled experiments, model selection, an interactive dashboard, and a laboratory deployment path built around OPC UA and Docker.

The project has two motivations. First, a quality-related measurement may not be continuously available as an online signal. Second, even when a recent laboratory result is available, anticipating its movement can provide an earlier indication of a change. These are different operational problems with different allowable inputs. Treating them as one task would overstate what a model can do.

The general objective was to create a transparent, reproducible advisory prototype for free-acid monitoring. The specific objectives were to understand the available historian extract, prepare time-respecting features, compare appropriate baselines and models, distinguish future forecasting from current-time virtual sensing, test robustness across chronological folds, and connect saved artifacts to a safe read-only streaming architecture.

The main contributions are: a documented feature contract with lags, rolling descriptors and process proxies; a target-anchored delta forecast that outperformed persistence on the held-out January period; a separate target-free virtual-sensor line of work; stability-oriented clustered Ridge experiments; and an advisory-only CSV-replay implementation with quality checks, warm-up, persistence and monitoring.

The report follows the engineering progression of the project. Chapter 2 provides the available company context. Chapters 3 to 7 cover the problem, data, methodology, results and interpretation. Chapters 8 to 10 describe deployment, implementation and limitations. The final chapters state the contribution, conclusions and realistic next work.

Chapter 2

Company Presentation

2.1 OCP Group

The OCP Group is a leading global producer of phosphate-based plant nutrition solutions and holds custody of approximately 68% of the world's estimated phosphate reserves.

Key Overview

- **Founded:** 1920 in Morocco as Office Chérifien des Phosphates.
- **Headquarters:** Casablanca, Morocco.
- **Core Activities:** Mining phosphate rock, producing phosphoric acid, and manufacturing customized fertilizers for global agriculture.
- **Global Reach:** Serves clients across five continents with an integrated value chain spanning extraction, processing, and shipping.
- **Recent Performance:** Reported a revenue of MAD 48.4 billion for the first half of 2026, maintaining a resilient EBITDA margin of 28% despite global market challenges and high sulfur costs.

2.2 Industrial activities relevant to the project

The project concerns the conversion of production measurements into information useful for process monitoring. The data dictionary identifies flows, temperatures, pressures, levels, recycle-related measurements and a free-acid target. These are characteristic of a production environment in which material feed, thermal conditions, slurry handling and drying-related signals are observed over time. The model does not replace process control. It produces an advisory estimate or forecast from read-only measurements.

2.3 Relevant technical roles

The implementation crosses several engineering roles. Process and production teams validate signal meaning, units, stream pairing, operating events and acceptable error. Automation and control teams own PLC/DCS logic, safety functions and industrial tags. Industrial IT and OT teams govern the OPC UA endpoint, certificates, host placement and firewall path. Data and software engineers maintain the feature contract, models, containers, storage and dashboard. Maintenance contributes sensor and communications context. The repository explicitly leaves several of these confirmations pending, which is appropriate for a staged industrial prototype.

Chapter 3

Internship Context and Problem Definition

3.1 TSP monitoring context

The confirmed target is SLURRY_FREE_ACID. Consequently, the project separates three contracts: direct future-level prediction, target-anchored forecasting and target-free current-time estimation. This separation is central to the validity of the conclusions.

Direct future-level prediction uses process information available at time t to estimate y_{t+h} . Target-anchored forecasting is allowed to use a fresh current laboratory result y_t and estimates its future change. Target-free virtual sensing estimates the present value from process variables only. The latter is not an early-warning forecast.

Table 3.1: Operational prediction contracts evaluated in the project.

Contract	Inputs	Output
Direct level	Process measurements at or before t	$\hat{y}_{t+h}$
Target anchored	Process information and fresh y_t	$\hat{y}_{t+h}$ through predicted change
Target free	Process information only	Current estimate $\hat{y}_t$

Table 3.1 prevents a common industrial error: presenting a current estimate as a future warning, or using a laboratory value as though it were a live sensor. The constraints also include time ordering, missing data, unknown unit semantics for several tags, limited data coverage, and the requirement that a model failure must not affect plant control.

3.2 Evolution of the work

The work evolved from exploration to feature preparation, direct benchmarks, delta forecasting, a target-free virtual sensor, robustness research, dashboard development and a replay-based

integration prototype. The final deployment code retains the target-free five-cluster Ridge candidate as a stability-oriented model. The target-anchored Ridge delta model remains a separate configurable mode. Neither is presented as a closed-loop controller or as fully validated on future production periods.

Chapter 4

Data Understanding and Preparation

4.1 Data source and scope

The notebooks use a confidential historian export named `tsp_1min.csv`. The raw file is intentionally absent from the repository, but generated audit artifacts record 44,640 one-minute rows from 1 to 31 January 2026. It contains 21 numeric process measurements and the target. There are 308 missing target values. Data were sorted by the `Date` timestamp, and all evaluation splits preserve chronology.

4.2 Variables and measurement semantics

Table 4.1 summarizes the principal variable families. Descriptions are taken from the variable registry, which marks several units and physical interpretations as provisional. The report consequently uses these signals as documented measurements rather than asserting unverified chemical or control meanings.

Table 4.1: Principal input families in the feature registry.

Family	Examples	Registry status
Production and flows	Production balance, acid lines, ground phosphate, slurry, fuel, wash liquid, steam, recycle	Several units and controllability claims require confirmation
Thermal measurements	Steam, attack tank, passage tank, slurry, dryer outlet, hot air and brick temperatures	Temperature names are documented; location or units may remain pending
Pressure and state	Steam pressure, dryer depression, passage-tank level	Units confirmed
Target	SLURRY_FREE_ACID	Laboratory or quality measurement, forbidden from target-free inputs

4.3 Data quality and preprocessing

The exploratory notebook examined timestamp order, cadence, duplicates, missingness, descriptive statistics, target extremes, correlations and time evolution. The target had 308 missing values and showed unusually extreme observations. For example, the observed minimum and maximum were 3.392 and 52.671, while the central distribution was concentrated near 11.3. These observations motivated diagnostics and conservative preprocessing, not automatic removal of all unusual values.

The structural warm-up removes the first 30 rows because required lagged and rolling history does not exist. Later isolated sensor gaps remain part of the chronology. The pipeline adds missingness indicators and uses training-fitted median imputation. Process-derived values use an $\epsilon = 10^{-6}$ denominator guard, replace infinities with missing values, and are clipped to training-period 0.1th and 99.9th percentiles. These rules prevent unstable ratios from dominating a model while avoiding validation or test fitting.

4.4 Feature engineering

For each process source, accepted temporal features include current value, selected lags, 10- and 30-minute rolling means, a 10-minute rolling standard deviation, a five-minute difference and a 10-minute rolling slope. Lags were selected from 1, 5, 10, 15, 30 and 60 minutes using training-only association, mutual information, stability across chronological blocks and a limited process-response prior. At most three lags were retained per source.

The implementation also creates total acid flow, acid-flow imbalance, acid-to-phosphate ratio, total slurry volumetric flow, a slurry-density proxy, thermal gradients, a steam-condition proxy, drying and recycle proxies, and selected flow coefficients of variation. Only the two slurry volumetric components are confirmed additive in the registry. The other derived relationships are explicitly treated as proxies.

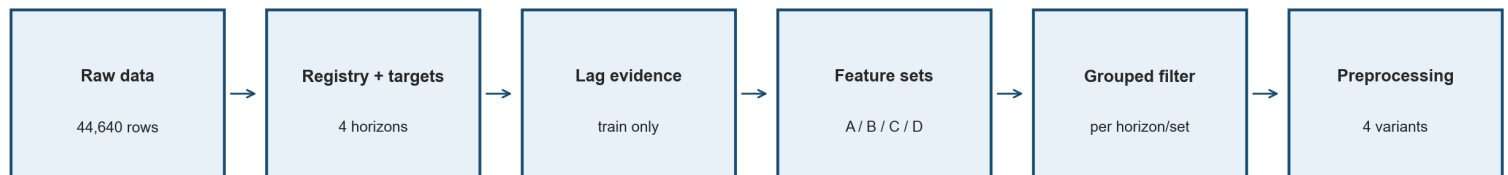


Figure 4.1: Leakage-controlled feature-preparation pipeline generated by Notebook 02.

Figure 4.1 shows the feature flow. Four hypotheses were tested: 21 raw variables, 162 retained temporal features, 58 process-aware features, and 178 full features after correlation filtering. Filtering was performed within source families using training-only absolute correlation above 0.98, while protected process-aware features and missingness indicators were retained.

Chapter 5

Machine Learning Methodology

5.1 Problem formulation

Let x_t denote the process information available at minute t and y_t the free-acid measurement. Direct forecasting learns $f_h(x_{\leq t}) \approx y_{t+h}$ for horizons $h \in \{1, 5, 10, 15\}$ minutes. Because a current observed level is very persistent over short periods, the project also evaluated delta forecasting:

$$\Delta y_{t,h} = y_{t+h} - y_t, \quad \hat{y}_{t+h} = y_t + \widehat{\Delta y_{t,h}}. \quad (5.1)$$

Equation 5.1 focuses the model on future movement rather than reproducing a known current level. It is valid only when the laboratory target is available and fresh. Target-free virtual sensing instead learns $g(x_{\leq t}) \approx y_t$ without any current or historical target feature.

Performance is reported with MAE and RMSE:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (5.2)$$

RMSE gives additional weight to larger errors, which is useful for identifying unstable periods, while MAE is readily interpretable as average absolute deviation.

5.2 Chronological validation

The baseline split is the first 70% for training, the next 15% for validation and the final 15% for test. Random shuffling was not used because adjacent one-minute samples are dependent and could make near-identical states appear in training and test. Preprocessing, lag selection, clipping limits, correlation filtering, hyperparameters and blend weights were learned without test fitting.

Later stability work used four expanding chronological blocks. For each fold, imputation, scaling, clustering and local experts were fitted on the preceding rows only. This produces a distribution of fold errors rather than a single validation number, allowing mean performance, variability

and worst-block behaviour to influence selection.

5.3 Models investigated

Ridge regression was retained as a regularized linear reference for correlated engineered features [1]. Extra Trees and Histogram Gradient Boosting were used in the direct benchmark to test nonlinear interactions. The target-free study additionally considered PLS [2], KNN, XGBoost or LightGBM [3] where available, and validation-weighted blends. The final Notebook 05 target-free baseline was a 50/50 Ridge-LightGBM blend. Chronological validation follows the need to respect serial dependence in time-series evaluation [4].

The robustness notebooks compared global blends, KMeans-partitioned local Ridge experts, a global/local hybrid and JITL variants using PCA or PLS representation with nearest neighbours. KMeans clusters are statistical feature-space groups, not confirmed operating regimes. JITL did not produce a selection improvement over the global blend.

Chapter 6

Experimental Results and Model Selection

6.1 Direct future-level benchmark

The first benchmark used process features to predict the absolute future target. Validation selected full-set Ridge at all horizons, but it did not beat the operational persistence baseline $\hat{y}_{t+h} = y_t$. Table 6.1 reports the untouched test result. This negative result was decisive because it showed that matching the current level is easier than predicting its movement.

Table 6.1: Direct future-level Ridge results on the held-out January period.

Horizon	MAE	RMSE	RMSE change versus persistence
1 min	0.269	0.332	-67.9%
5 min	0.275	0.338	-35.1%
10 min	0.280	0.344	-22.6%
15 min	0.304	0.378	-27.5%

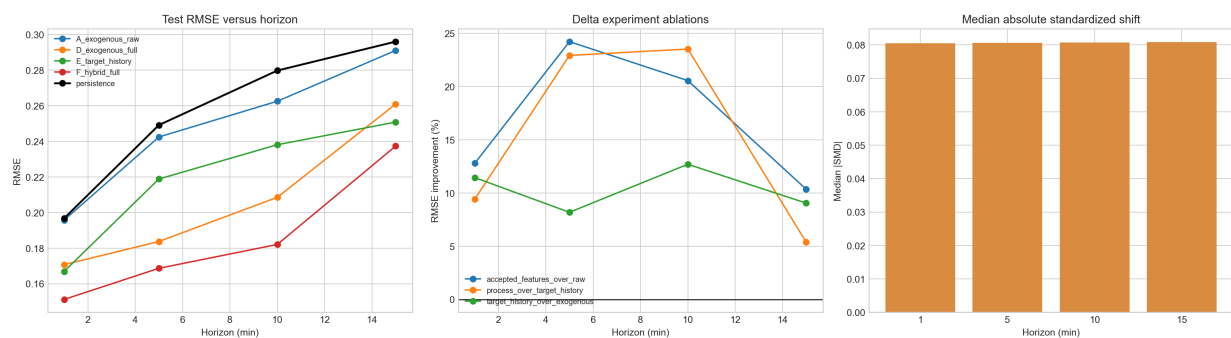
6.2 Target-anchored delta forecast

Ridge and Extra Trees received comparable tuning budgets on raw process, full exogenous, target-history and hybrid feature sets. Validation selected a hybrid Ridge model with $\alpha = 10$ at every horizon. Table 6.2 shows that the reconstructed delta forecast improved RMSE over persistence at all tested horizons. The 10-minute setting had the largest relative gain, while the one-minute setting had the smallest absolute error. The operational horizon would still require a decision about reaction time and acceptable false or missed warnings.

Table 6.2: Target-anchored delta-Ridge test performance after future-level reconstruction.

Horizon	MAE	RMSE	R^2	RMSE improvement
1 min	0.102	0.151	0.889	23.2%
5 min	0.113	0.169	0.862	32.3%
10 min	0.126	0.182	0.839	34.9%
15 min	0.176	0.237	0.726	19.8%

Figure 6.1 combines delta performance, ablation and distribution-shift diagnostics. Accepted engineered features improved raw process input, while target history and process information supplied complementary predictive association. Candidate transition weighting slightly helped some local errors but worsened total RMSE, so it was not retained. A logistic transition classifier also remained diagnostic only because precision was low and false-warning rates were too high for an operational alarm.

**Figure 6.1:** Delta forecasting performance, ablation and distribution-shift diagnostic.

6.3 Target-free virtual sensor and stability research

The target-free study forbade current target, target history and future target columns. The validation-selected 50/50 Ridge-LightGBM blend achieved held-out MAE 0.249, RMSE 0.301 and $R^2 = 0.558$, compared with RMSE 0.454 for the training-mean baseline. Its empirical 90% interval covered 89.7% of January test points with mean width 1.015. It is an advisory diagnostic because the interval was calibrated from individual-Ridge validation residuals while the point estimate is a blend.

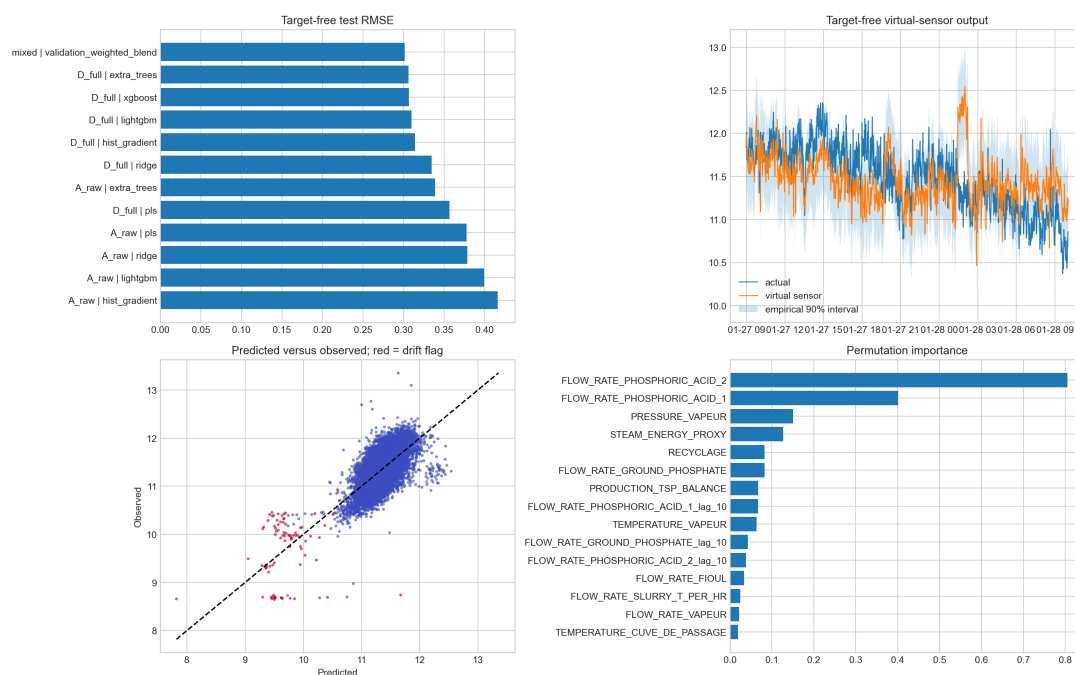


Figure 6.2: Target-free virtual-sensor comparison and diagnostics from Notebook 05.

Figure 6.2 illustrates the selected virtual-sensor investigation. Subsequently, four-fold stability research found that five-cluster Ridge had mean RMSE 0.3280, standard deviation 0.0396 and worst-fold RMSE 0.3522. This was less favorable in mean RMSE than the 70/30 global Ridge-LightGBM blend at 0.3191, but materially steadier than its 0.0762 standard deviation and 0.4236 worst fold. The target-free runtime loads the five-cluster Ridge candidate for this stability-first reason.

The later 80% local and 20% global hybrid achieved mean RMSE 0.3213, standard deviation 0.0413 and worst RMSE 0.3548. It remained a research candidate. JITL variants did not supersede the global blend. Since January had already been used during the sequence of experiments, retrospective refits cannot be presented as new test evidence. A later labeled month is needed before production-model replacement.

Chapter 7

Model Interpretability

The project did not run SHAP analysis. The notebooks state this explicitly and instead use model-agnostic permutation importance, grouped source and feature-family importance, Ridge coefficient inspection, and importance-stability diagnostics. This is a useful methodological distinction because it avoids claiming a SHAP result that was never generated.

Permutation importance measures the deterioration in prediction performance when a feature is disrupted. Ridge coefficients describe the fitted linear contribution after scaling. Neither method establishes process causality, controllability, or a safe intervention, a general limitation of model-agnostic explanation methods [5]. In this project, importance was used to check whether feature families and accepted process variables contributed predictive association, and to guide discussions with process engineers.

The dashboard and notebooks also inspect candidate transitions using a multivariate normalized process-change score. The score is a diagnostic of joint sensor movement. It is not target change, an identified startup or shutdown, a causal attribution, or a model output. This distinction is important because the available data does not include validated event labels.

Figure 7.1 presents the existing benchmark interpretation material. It should be read together with the feature registry: some variables have provisional units, uncertain stream meanings, or unconfirmed controllability. The correct use of this analysis is to prioritize validation questions, not to recommend changing a process variable.

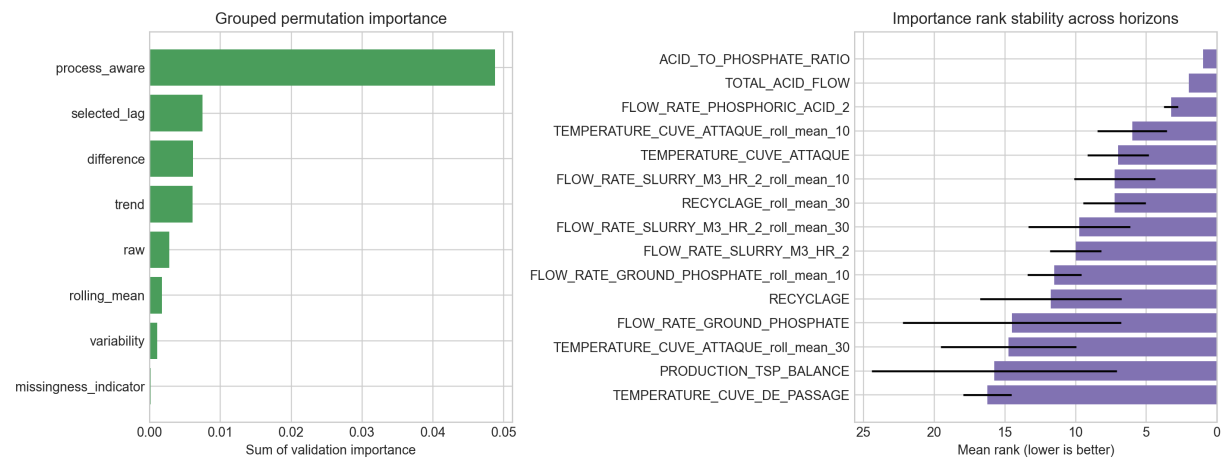


Figure 7.1: Grouped predictive importance and horizon stability from the direct-benchmark analysis.

Chapter 8

Deployment Architecture

8.1 Architecture and safety boundary

The implementation is a staged, advisory-only architecture. Figure 8.1 summarizes the intended information path. Sensors and laboratory information are organized by PLC/DCS systems. Kepware is the intended industrial gateway: it communicates southbound with approved plant protocols and exposes configured tags northbound through OPC UA [6]. The Python inference service reads those tags, prepares features and writes predictions to local audit storage and the dashboard. The optional advisory OPC namespace is disabled by default.

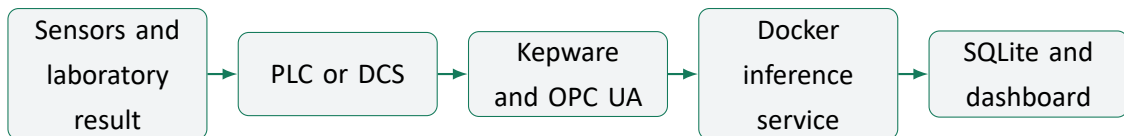


Figure 8.1: Advisory information flow implemented and planned for industrial integration.

The model has no path to modify PLC/DCS setpoints, controller modes, interlocks, permissives, valve commands, pumps or safety logic. A service, network, database or dashboard failure must leave control independent. A real connection still requires approved endpoint details, certificates, security policy, Node IDs, firewall routes, quality semantics and IT/OT authorization. The repository uses placeholder mapping values for this purpose.

8.2 Containers, services and configuration

Docker Compose defines two services on an isolated bridge network: an OPC UA CSV-replay simulator and a Python inference service. Both use Python 3.11 slim images. The inference image installs required packages and `libgomp1` for the LightGBM runtime, runs a health check, and mounts configuration, model artifacts, schema and runtime output. Runtime settings, including mode, horizon, endpoint and advisory-write setting, can be supplied through environment variables. Node IDs are externalized in `opc_tags.yaml`, not embedded in Python.

SQLite is the authoritative Stage 1 audit store, with JSONL and latest-JSON mirrors for convenient inspection. It records service runs, predictions and status events. WAL mode permits a read-only dashboard to query history while the service writes. This is appropriate for a local demonstration with one writer; it is not claimed as a replacement for a plant historian, central retention policy or backup design.

Chapter 9

Real-Time Processing and Deployment Implementation

9.1 Stage 1 CSV replay

Stage 1 reproduces a streaming environment without plant access. The simulator reads the confidential historical CSV, publishes row values as OPC UA tags, sends source timestamp and then writes a sequence number as a commit marker. It can update the simulated laboratory target every 60 rows. This replay validates tag mapping, coherent snapshot acquisition, buffering, model loading, storage and restart behavior. It is a laboratory simulation, not a real PLC/DCS deployment.

9.2 Streaming pipeline and quality controls

The inference service subscribes to the sequence tag rather than polling every sensor repeatedly. It reads sequence, input tags and sequence again, accepting a snapshot only when the sequence values agree. This avoids a mixed snapshot spanning two replay rows. Accepted snapshots pass through a quality shield that checks OPC quality, timestamp order and cadence, impossible values and the share of missing required inputs.

Isolated missing cells are accepted as `IMPUTED`, then handled by the saved training-fitted imputer and missingness indicators. A snapshot with more than 25% missing required inputs, bad or uncertain quality, or an invalid chronology is rejected as `DATA_ERROR`. The service never fits preprocessing during live inference.

9.3 Warm-up and supported modes

The maximum feature lookback is 30 minutes. A cold start requires the current row plus those 30 previous rows, hence 31 accepted rows before valid inference. Until then the service reports

WARMING_UP; it does not fabricate history. A production improvement would be historian backfill, followed by carefully validated persistent-buffer restoration.

The `target_free` mode estimates the current target from process signals. The `target_anchored` mode requires the current laboratory value and its timestamp, then produces 1-, 5-, 10- or 15-minute delta forecasts. The configuration rejects a manual target that is missing, future-dated or older than 120 minutes. This protects the system from treating an outdated laboratory result as a real-time analyzer.

9.4 Heartbeat, watchdogs and recovery

The service refreshes health information every five seconds, including periods with no new source row. The desktop dashboard reads health every two seconds and can use a 30-, 90- or 180-second watchdog threshold, or disable the alert. This distinguishes an expected quiet process from stale service information. The dashboard reads SQLite trend history every ten seconds and uses deterministic connection closing to avoid Windows locking issues.

The OPC sequence watchdog polls the sequence node while the source is quiet, which can reveal a broken subscription or gateway reset. Connection closure or sequence reset clears incompatible buffer history, enters `DISCONNECTED`, and follows bounded reconnect attempts. After restoration, a new `run_id` is created, the buffer warms up again, and old-run predictions are not presented as current-run values. The simulator has an acknowledgement mechanism that waits for inference to process each sequence. This is intentionally laboratory-only flow control and must be disabled with a real Kepware or plant source.

Table 9.1: Operational states exposed by the inference service.

State	Meaning
STARTING or CONNECTED	The service has started or has an OPC session but no model-ready history.
WARMING_UP	Accepted history is accumulating toward the 31-row requirement.
RUNNING	A recent accepted snapshot has produced a prediction.
DATA_ERROR	A material data-quality or chronology rule blocked inference.
DISCONNECTED	OPC communication failed or reset; reconnection is attempted.

Table 9.1 makes the dashboard useful as a monitoring surface without suggesting that it controls production. The dashboard presents current prediction and quality cards, a trend view, imputed-input markers, delta versus the previous prediction and a heartbeat watchdog. It reads runtime output only.

Chapter 10

Discussion

The strongest evidence in the project is the target-anchored delta formulation on the held-out January period. It improved over persistence for all four horizons, whereas direct level prediction did not. This is a meaningful modeling result, but its use is conditional on a recent and correctly timestamped laboratory value. The target-free virtual sensor solves a different problem and should not be described as a substitute for a future warning.

The later five-cluster Ridge decision prioritizes repeatability across four chronological folds, not the lowest average fold RMSE. That choice is reasonable for a prototype designed to avoid unstable behavior, but it remains a selection from one month. Its statistical clusters cannot be equated with confirmed process regimes until they are checked against operator records such as startup, shutdown, maintenance, cleaning, grade changes and disturbances.

Several limitations constrain interpretation. The dataset spans only January 2026; sensor semantics and units are incomplete for many tags; free acid is configured as manual laboratory data; and no variables are confirmed controllable. Empirical uncertainty and drift flags identify cases requiring caution, but they cannot prove correctness under a new operating distribution. The deployment path was tested in CSV replay, not against a secured plant endpoint. The prototype also does not establish production hardware, long-term retention, access control, backup, or model-governance arrangements.

These limitations explain the architecture choices. Quality shields, chronological buffers, warm-up, status events, heartbeat, reconnection and read-only storage are not optional interface details. They preserve the distinction between an offline model result and a trustworthy advisory service. The correct next industrial step is controlled shadow operation, not automatic action.

Chapter 11

Contribution and Lessons Learned

This internship produced a reusable engineering chain from historian-style data to an advisory prototype. On the data-science side, the work demonstrated why time ordering, a persistence baseline and leakage controls are essential for one-minute process data. It also showed that a change target can be more appropriate than absolute-level prediction when a current measurement is highly persistent.

On the software side, the project converted notebook logic into saved artifacts, a feature builder that reproduces the accepted feature contract, configuration-driven OPC mapping, tests for buffer, quality, storage and resilience behavior, containers, and a read-only dashboard. The work reinforced that industrial ML depends on traceability of feature meaning, model inputs, health state and output history.

The most important practical lesson is scope discipline. A simulator confirms software integration behavior but not production validity. Predictive importance does not prove causality. A model prediction does not authorize a process intervention. These distinctions make the delivered prototype more useful because they identify the exact information and validation needed before expanded deployment.

Chapter 12

Conclusion and Future Work

The internship investigated free-acid monitoring for a TSP process using one month of one-minute production data. The project moved from exploration and feature engineering to controlled benchmarks, target-anchored delta forecasting, target-free virtual sensing, stability research, dashboards and a Docker/OPC UA laboratory prototype.

The main forecasting result is a hybrid-feature Ridge delta model that beat persistence at 1, 5, 10 and 15 minutes on the held-out January period. A separate target-free Ridge-LightGBM virtual sensor outperformed a constant baseline, while five-cluster Ridge was retained by the deployment runtime as a stability-oriented candidate after chronological-fold analysis. The report does not combine these results: they answer different questions and have different evidence levels.

Future work should first obtain a later labeled production period and evaluate frozen candidates without retuning. It should then perform rolling-origin multi-month validation, verify units and proxy assumptions with process engineers, define laboratory freshness and warning costs, calibrate drift and uncertainty on independent residuals, and validate clusters against operational records. On the deployment side, the next milestones are secured Kepware integration, historian backfill, retention and backup policy, longer read-only shadow operation and operator feedback. Any move beyond advisory use would require separate governance and control-system approval.

Bibliography

- [1] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.
- [2] S. Wold, M. Sjostrom, and L. Eriksson, "Pls-regression: A basic tool of chemometrics," *Chemometrics and Intelligent Laboratory Systems*, vol. 58, no. 2, pp. 109–130, 2001.
- [3] G. Ke *et al.*, "Lightgbm: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] C. Bergmeir, R. J. Hyndman, and B. Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction," *Computational Statistics and Data Analysis*, vol. 120, pp. 70–83, 2018.
- [5] C. Molnar *et al.*, "General pitfalls of model-agnostic interpretation methods for machine learning models," *arXiv preprint arXiv:2007.04131*, 2020.
- [6] "Opc unified architecture specification," 2024.

Appendix A

Evidence and Deployment Scope

A.1 Notebook and implementation evidence

The report is based on the repository notebooks 01 through 08, their generated CSV artifacts, the existing technical reports, and the deployment source code. The following progression was retained: exploratory analysis, prepared feature sets, direct benchmark, delta and transition analysis, target-free virtual sensing, robust candidate comparison, global and local hybrid research and JITL research. The source artifacts remain the detailed audit record for all rows, folds, coefficients, predictions and configuration values summarized here.

A.2 Terms that require supervisor confirmation

Before production shadow mode, the following repository items require owner validation: exact engineering units and material basis for several flow and thermal signals; recycle stream meaning; correspondence between slurry mass and combined volumetric streams; actual event labels; source timestamp and freshness policy for laboratory free acid; DCS vendor and endpoint; Node IDs; OPC UA certificates and security policy; network route; retention and access policy; and the decision cost of errors and warnings.

A.3 Software verification scope

The deployment repository includes unit tests for storage, quality checks, buffer chronology, configuration, resilience and dashboard data access. These checks exercise software behavior. They do not validate the chemical process, sensor calibration, industrial cybersecurity, plant availability, or the business usefulness of an advisory prediction.

Machine Learning for TSP Free-Acid Monitoring

Final-year engineering internship report

Abstract. This report documents a machine-learning project for estimating and forecasting `SLURRY_FREE_ACID` from one-minute TSP process data. It presents leakage-controlled feature engineering, target-anchored delta forecasting, target-free virtual sensing, stability-oriented model selection, and an advisory-only Docker and OPC UA replay prototype. The work is limited to one month of data and a laboratory replay environment. Further multi-month validation and read-only industrial shadow operation are required before operational adoption.

Ikram Benfella